

Statistics Lecture 2

Parameter estimation

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Connectivity

Outline

1. Intro. to parameter estimation
2. Estimating mean and variance
3. Maximum likelihood estimation (MLE)

Intro. to parameter estimation

Parametric estimation

We observe a sample $X_1, X_2, \dots, X_n$

Each X_i is a random variable with **known distribution** F with parameter θ

Parameter θ is a fixed value and not an RV

Estimate of θ from the sample with n points using function h

$$\hat{\theta}_n = h(X_1, X_2, \dots, X_n)$$

Our statistic $\hat{\theta}_n$ IS a RV because it depends on the sampled data

What are the properties of $\hat{\theta}_n$?

We hope that **the estimator** $\hat{\theta}_n$ is close to the real value of θ ...

... as frequently as possible

Non-parametric estimation

We don't need to know the type of distribution F

When compared to parametric estimation



Much more general



Much more difficult to do

We'll only work with parametric estimation

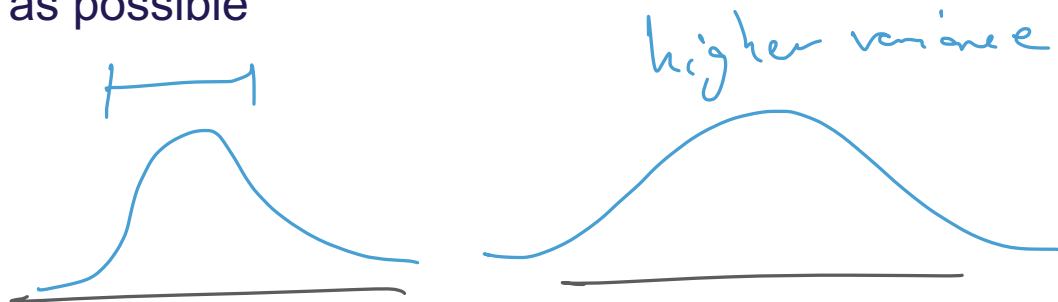
Metrics to evaluate estimators

An estimator should be as close as possible to the true value
This should happen as frequently as possible

What does this mean?

Metrics for evaluation:

- **Bias:** measures accuracy
- **Variance:** measures precision
- **Mean square error:** measures both accuracy and precision



Bias

Measure of how close the estimate is to the true value

Let $\hat{\theta}_n = h(X_1, X_2, \dots, X_n)$ be the estimator for parameter θ

The bias is defined as

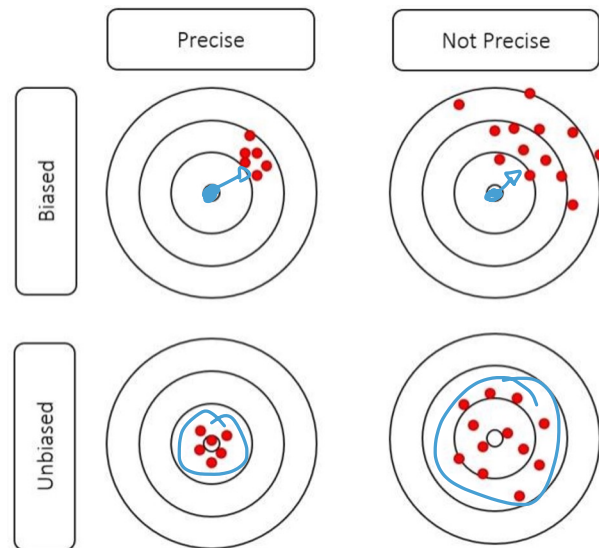
$$\text{bias}(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n) - \theta$$


Unbiased estimator:

An estimator with $\text{bias}(\hat{\theta}_n) = 0$

This is what we hope for

There are tricks to correct the bias



Variance and standard error

Measure of how precise the estimator is for different samples

Let $\hat{\theta}_n = h(X_1, X_2, \dots, X_n)$ be the estimator for parameter θ

The variance is

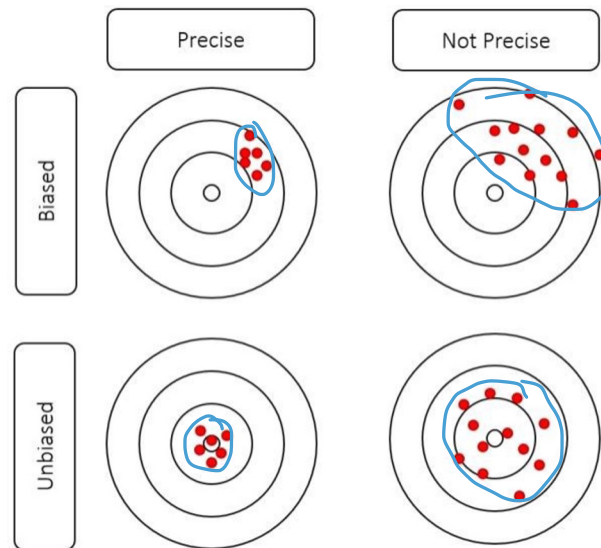
$$\text{var}(\hat{\theta}_n) = \mathbb{E} \left[\left(\hat{\theta}_n - \mathbb{E}(\hat{\theta}_n) \right)^2 \right]$$

The standard error is

$$\text{se}(\hat{\theta}_n) = \sqrt{\text{var}(\hat{\theta}_n)}$$

but $\text{se}(\hat{\theta}_n)$ depends on F and might be unknown

The estimated standard error is $\hat{\text{se}}(\hat{\theta}_n)$



The Mean Squared Error (MSE)

We define the **Mean Squared Error (MSE)** of an estimator as

$$\text{MSE} = \mathbb{E}[\hat{\theta}_n - \theta]^2 = \underbrace{\text{bias}^2(\hat{\theta}_n)} + \underbrace{\text{var}(\hat{\theta}_n)}$$

Combines the bias and variance

An estimator is consistent if $\underbrace{\text{bias}(\hat{\theta}_n)} \rightarrow 0$ and $\underbrace{\text{var}(\hat{\theta}_n)} \rightarrow 0$ as $\underbrace{n \rightarrow \infty}$

This means that $\underbrace{\hat{\theta}_n \xrightarrow{P} \theta}$

Exercise: Comparing estimators

Let $X_1, X_2, \dots, X_n$ be i.i.d. RVs with mean $\mathbb{E}[X_i] = \theta$ and variance $\text{var}(X_i) = \sigma^2$

Consider the following two estimators for θ

1. $\hat{\theta}_1 = X_1$

2. $\hat{\theta}_n = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ wins

Which one is a better estimator?

① $\text{bias}(\hat{\theta}_1) = \mathbb{E}[\hat{\theta}_1] - \theta = \theta - \theta = 0$ ② $\text{bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta$

$\text{var}(\hat{\theta}_1) = \sigma^2$

$\mathbb{E}[\hat{\theta}_n] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \theta$

$\text{var}(\hat{\theta}_n) = \text{var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{\sigma^2}{n}$

Estimating the mean and variance

Properties of the sample mean

If $X_1, X_2, \dots, X_n$ are random variables with $\mathbb{E}(X_i) = \mu$ and $\text{var}(X_i) = \sigma^2$

Central limit theorem:

$$\sum_{i=1}^n X_i \approx N(n\mu, n\sigma^2)$$

The sample mean is the random variable

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$
$$\mathbb{E}(\bar{X}_n) = \mu$$

The law of large numbers

$\bar{X}_n$ **converges in probability** to $\mu = \mathbb{E}(X_i)$ as $n \rightarrow \infty$

$$\bar{X}_n \xrightarrow{P} \mu = \mathbb{E}(X_i)$$

This means that, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \epsilon) = 0$$

So, $\bar{X}_n$ is close to μ with high probability if the sample size is large

Variance of the sample mean

If $X_1, X_2, \dots, X_n$ are random variables with $\mathbb{E}(X_i) = \mu$ **and** $\text{var}(X_i) = \sigma^2$

Central limit theorem:

$$\sum_{i=1}^n X_i \approx N(n\mu, n\sigma^2)$$

$$\text{var}\left(\sum_{i=1}^n a_i X_i + b\right) = \sum_{i=1}^n a_i^2 \text{var}(X_i)$$



$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

The variance of the sample mean is $\text{var}(\bar{X}_n) = \frac{\sigma^2}{n}$

Estimating the variance

Sample variance

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

$\text{dof} = 1$
dof: degrees of freedom

The term $n - 1$ is needed to have an unbiased estimate

$$\mathbb{E}[S_n^2] = \sigma^2$$

Continue 9:11

Evaluating estimators

Exercise cont.: Comparing estimators

1. $\hat{\theta}_1 = X_1$:

$$\text{bias}(\hat{\theta}_1) = \mathbb{E}[X_1] - \theta = 0 \text{ and } \text{var}(X_1) = \sigma^2$$

The estimator $\hat{\theta}_1 = X_1$ **is not consistent**

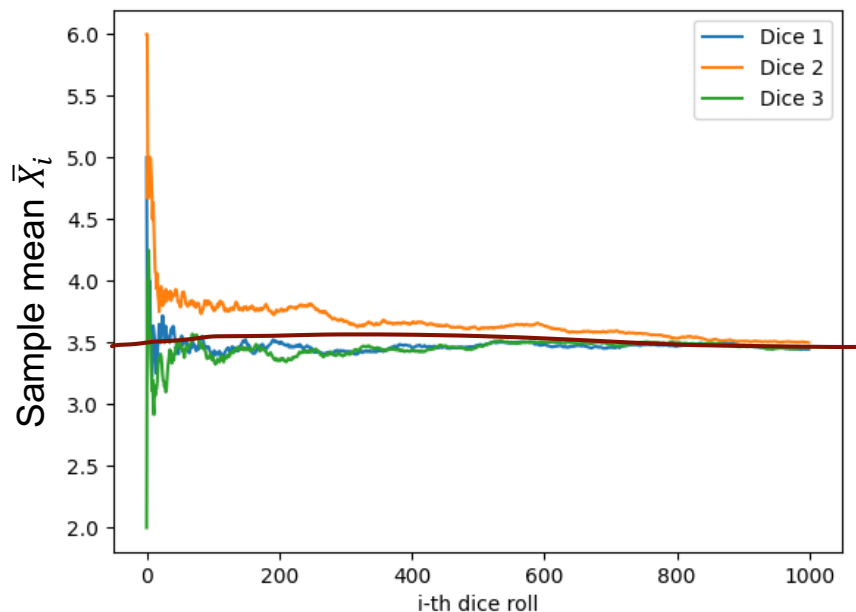
2. $\hat{\theta}_n = \bar{X}_n$:

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}[\bar{X}_n] - \theta = 0 \text{ and } \text{var}(\bar{X}_n) = \sigma^2/n$$

Since $\text{var}(\bar{X}_n) = \frac{\sigma^2}{n} \rightarrow 0$ as $n \rightarrow \infty$, **the estimator $\hat{\theta}_n = \bar{X}_n$ is consistent**

Dice example

We roll a dice $i = 1, 2, \dots, n$ times and compute the sample mean $\bar{X}_i$



Exercise: Bias for estimator in Bernoulli RVs

Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$. We don't know the real value of p
We have to create an estimate for p , defined as

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$$

$$\mathbb{E}(\hat{p}_n) = \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) = p$$

$$\frac{1}{n} \sum_{i=1}^n p = \frac{np}{n} = p$$

Is this estimator biased or not?

$$\text{bias}(\hat{p}_n) = \mathbb{E}(\hat{p}_n) - p = 0$$

Exercise: Variance of estimator for Bernoulli RVs

The **variance of a RV** is $\text{var}(X_i) = \mathbb{E}[(X_i - \mu_i)^2] = \mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2$

$$\text{var}\left(\sum_{i=1}^n a_i X_i + b\right) = \sum_{i=1}^n a_i^2 \text{var}(X_i)$$

The **variance** for our estimator is

$X_i \sim \text{Bernoulli}(p)$

$$\text{var}(\hat{p}_n) = \text{var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \boxed{p(1-p)} = \frac{p(1-p)}{n}$$

$\text{var}(X_i)$

Is this estimator consistent or not?

Yes it tends to 0 as $n \rightarrow \infty$

Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation (MLE)

The most common method for parametric estimation

The likelihood function for $X_1, X_2, \dots, X_n$ i.i.d RVs with PDF/pmf $f(x; \theta)$ is

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f(X_i; \theta)$$

Usually, we use the log-likelihood function is

$$\ell_n(\theta) = \log \mathcal{L}_n(\theta) = \sum_{i=1}^n \log f(X_i; \theta)$$

We treat the likelihood function as a function of parameter θ

Maximum Likelihood Estimator (MLE) $\hat{\theta}_n$: the value of θ that maximizes $\mathcal{L}_n(\theta)$ or $\ell_n(\theta)$

The MLE is typically consistent

MLE for discrete and continuous RVs

Suppose we observe the outcomes $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$

For discrete RVs, the likelihood function is the joint pmf

$$\mathcal{L}_n(\theta) = \mathcal{L}(x_1, x_2, \dots, x_n; \theta) = P_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; \theta)$$

For continuous RVs, the likelihood function is the joint PDF

$$\mathcal{L}_n(\theta) = \mathcal{L}(x_1, x_2, \dots, x_n; \theta) = f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; \theta)$$

Exercise: MLE for Bernoulli RV pt. 1

Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$ with pmf

$$f(x; p) = p^x (1 - p)^{1-x},$$

$$x \in \{0, 1\}$$

$$P_r(x=0) = (1-p)$$

$$P_r(x=1) = p$$

The likelihood function is

$$\mathcal{L}_n(p) = \prod_{i=1}^n f(X_i; p) = \prod_{i=1}^n p^{X_i} (1 - p)^{1-X_i}$$

If we define $\underline{X} = \sum_{i=1}^n X_i$ we get the likelihood and log-likelihood functions:

$$\mathcal{L}_n(p) = p^X (1 - p)^{n-X}$$

log-likelihood $\underline{\ell}_n(p) = \log(p^X (1 - p)^{n-X}) = \log(p^X) + \log((1-p)^{n-X})$
 $= X \log(p) + (n-X) \log(1-p)$

Which one should we work with?

Exercise: MLE for Bernoulli RV pt. 2

How do we find the value of p that maximizes the likelihood $\mathcal{L}_n(p)$ or $\ell_n(p)$?

1. Take the derivative with respect to p and make it equal to 0

$$\frac{d\mathcal{L}_n(p)}{dp} = Xp^{X-1}(1-p)^{n-X} - (n-X)(1-p)^{n-X-1}p^X = 0$$

$$\frac{d\ell_n(p)}{dp} = \frac{X}{p} - \frac{n-X}{(1-p)} = 0$$

2. Solve for p

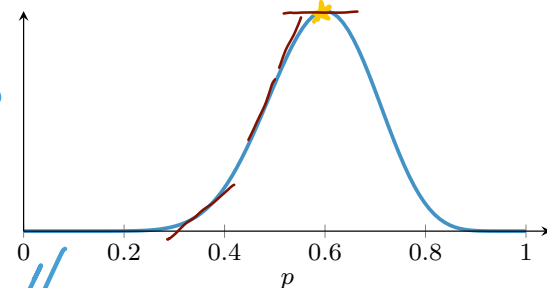
$$\hat{p}_n = \frac{\sum_{i=1}^n x_i}{n} = \bar{X}_n$$

$$\Rightarrow \frac{X}{p} - X = n - X \Rightarrow \frac{X}{p} = n \Rightarrow p = \frac{X}{n}$$

$$\frac{X}{p} = \frac{n-X}{1-p}$$
$$\frac{(1-p)X}{p} = n - X \Rightarrow$$

Likelihood function

$n = 20$ and $X = 12$



Finally, an exercise with numbers pt. 1

There's a bag with 3 balls. Each ball is either orange or green.
The number of green balls is $\theta \in \{0, 1, 2, 3\}$.

We estimate θ by grabbing and putting back a ball 4 times

This is a **selection with replacement**

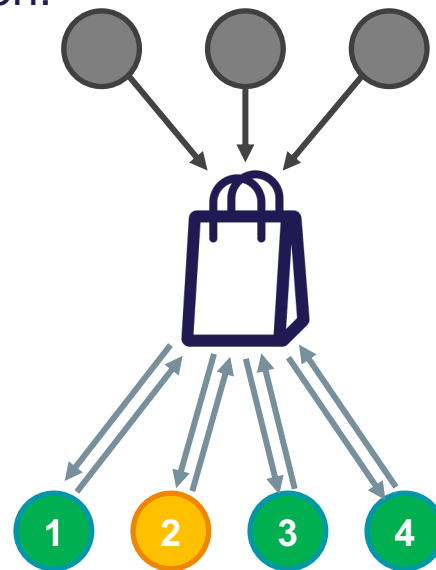
Let X_1, X_2, X_3, X_4 be the RVs of the color of the i -th ball

$$X_i = \begin{cases} 1, & \text{if the ball is green} \\ 0, & \text{if the ball is orange} \end{cases}$$

Then, $X_i \sim \text{Bernoulli}\left(\frac{\theta}{3}\right)$

The outcomes are $x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 1$

That is, 3 out of 4 of the balls are green



Finally, an exercise with numbers pt. 2

Option 1. For each possible value of θ , find $P_{X_1, X_2, X_3, X_4}(1, 0, 1, 1; \theta)$

1

2

3

4

$$P_{X_1, X_2, X_3, X_4}(1, 0, 1, 1; \theta = 0) = \prod_{i=1}^4 P(X_i = x_i; \theta = 0) = 0 \times 1 \times 0 \times 0 = 0$$

$$P_{X_1, X_2, X_3, X_4}(1, 0, 1, 1; \theta = 1) = \prod_{i=1}^4 P(X_i = x_i; \theta = 1) = \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{2}{81} = 0.0247$$

$$P_{X_1, X_2, X_3, X_4}(1, 0, 1, 1; \theta = 2) = \prod_{i=1}^4 P(X_i = x_i; \theta = 2) = \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{81} = 0.0987$$

$$P_{X_1, X_2, X_3, X_4}(1, 0, 1, 1; \theta = 3) = 1 \times 0 \times 1 \times 1 = 0$$

Finally, an exercise with numbers pt. 3

Option 2. Calculate the MLE for θ with n experiments given $p = \theta/3$.

Previously, we defined $\ell_n(p) = \log(p^X (1-p)^{n-X}) = X \log p + (n-X) \log(1-p)$

$$\ell_n(\theta) = \log \left(\left(\frac{\theta}{3} \right)^X \left(1 - \frac{\theta}{3} \right)^{n-X} \right) = X(\log \theta - \log 3) + (n-X) \log \left(1 - \frac{\theta}{3} \right)$$

$$\frac{d\ell_n(\theta)}{d\theta} = \frac{X}{\theta} - \frac{n-X}{3 \left(1 - \frac{\theta}{3} \right)} = 0$$

$$X(3 - \theta) = \theta(n - X) \Rightarrow 3X - \theta X = n\theta - \theta X \Rightarrow \theta = \frac{3X}{n} = 3\bar{X}_n$$

Finally, an exercise with numbers pt. 4

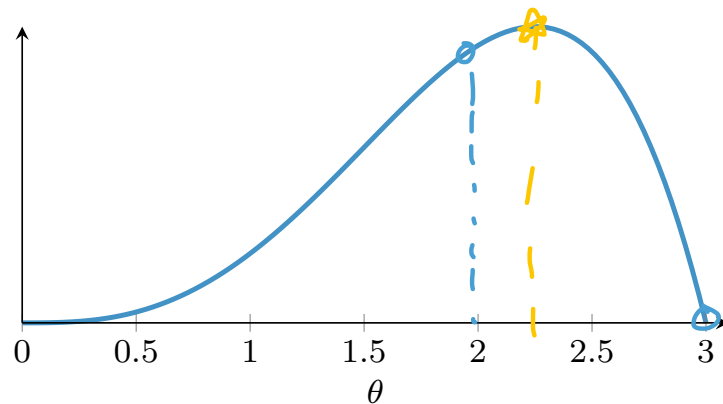
For $n = 4$ and $X = 3$ we got $\bar{X}_4 = 3/4$ and the MLE becomes

$$\hat{\theta}_4 = 3\bar{X}_4 = 3 \times 3/4 = 9/4 = 2.25$$

Since $\hat{\theta}_n$ must be an integer, we take the integer with the highest value: $\hat{\theta}_4 = 2$



$$\begin{aligned}\ell_n(\theta) &= X(\log \theta - \log 3) + (n - X) \log \left(1 - \frac{\theta}{3}\right) \\ &= 3(\log \theta - \log 3) + (4 - 3) \log \left(1 - \frac{\theta}{3}\right)\end{aligned}$$



Cramér-Rao Lower Bound (CRLB)

This is a fundamental limit to the variance of unbiased estimators

$$\text{var}(\hat{\theta}_n) \geq \frac{1}{I_n(\theta)}$$

where $I_n(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ell_n(\theta) \right)^2 \right]$ is the Fisher information for sample size n .

Important: $I_n(\theta) = n I_1(\theta)$

Example: For a Bernoulli RV and $n = 1$ we know that $I_1(p) = \frac{1}{p(1-p)}$.

So, $\text{var}(\hat{p}_n) \geq \frac{p(1-p)}{n}$

$\text{var}(x_i)$

Summary

Summary

Sampling is essential for estimation

In **parametric estimation**, we know the distribution but not the parameter

- The estimator is a RV that takes a value based on the sample

In **non-parametric estimation**, we don't need to know the distribution

There can be an infinite number of estimators

We find the best ones using **metrics: bias, variance, and MSE**

If the MSE goes to 0, the estimator is **consistent**

The Maximum Likelihood Estimator (MLE) is consistent

Appendix

Useful derivative formulas and rules

$$\frac{da^x}{dx} = \log(a)a^x$$

$$\frac{d \log_a(x)}{dx} = \frac{1}{x \log(a)}$$

$$\frac{d \log(x)}{dx} = \frac{1}{x}$$

Product rule: $h(x) = f(x)g(x)$

$$h'(x) = f'(x)g(x) + f(x)g'(x)$$

Quotient rule: $h(x) = f(x)/g(x)$

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

Chain rule: $h(x) = f(g(x))$

$$h'(x) = f'(g(x))g'(x)$$

Derivations for Bernoulli MLE (log-likelihood)

$$\frac{d\ell_n(p)}{dp} = \frac{X}{p} + \frac{n-X}{(1-p)}(-1) = \frac{X}{p} - \frac{n-X}{(1-p)} = 0$$

$$\frac{X}{p} = \frac{n-X}{(1-p)}$$

$$X(1-p) = p(n-X) \Rightarrow X - Xp = np - Xp$$

$$X = np \Rightarrow p = \hat{p}_n = \frac{X}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

Derivations for unbiased variance estimator (pt. 1)

Our sample variance estimator is

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

Is it biased?

Recall the formula: $\text{bias}(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n) - \theta$

$$\text{bias}(S_n^2) = \mathbb{E}(S_n^2) - \sigma^2$$

$$\mathbb{E}(S_n^2) = \mathbb{E} \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \right] = \frac{1}{n-1} \sum_{i=1}^n \mathbb{E}(X_i^2) - 2\mathbb{E} \left(\bar{X}_n \sum_{i=1}^n X_i \right) + \sum_{i=1}^n \mathbb{E}(\bar{X}_n^2)$$

Derivations for unbiased variance estimator (pt. 2)

$$\begin{aligned}\mathbb{E}(S_n^2) &= \mathbb{E}\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2\right] = \frac{1}{n-1} \left[\sum_{i=1}^n \mathbb{E}(X_i^2) - 2\mathbb{E}\left(\bar{X}_n \sum_{i=1}^n X_i\right) + \sum_{i=1}^n \mathbb{E}(\bar{X}_n^2) \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n \text{Var}(X_i) + \mathbb{E}(X_i) - 2\mathbb{E}(n\bar{X}_n^2) + \sum_{i=1}^n \text{var}(\bar{X}_n) + \mathbb{E}(\bar{X}_n) \right] \\ \mathbb{E}(S_n^2) &= \frac{1}{n-1} \left[n\sigma^2 + n\mu - 2n\left(\frac{\sigma^2}{n} + \mu\right) + \sigma^2 + n\mu \right] = \frac{\sigma^2(n-1)}{n-1} = \sigma^2\end{aligned}$$

Since $\text{bias}(S_n^2) = \mathbb{E}(S_n^2) - \sigma^2 = 0$, **this estimator is unbiased.**

If the denominator were n , the bias would be $-\frac{\sigma^2}{n}$